ModelIt!

Student Activity Pack

Phase Changes: When Matter Transforms

Same Stuff, Different State — and Nothing Is Lost

Grade 05L2 | NGSS 5-PS1-2, 5-PS1-4 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What happens to ice when you leave it on a counter? Where does it go?

2 When you boil water and see steam, did the water disappear? Explain your thinking.

3 Draw what you think molecules look like in ice, liquid water, and steam.

4 If you weigh a glass of water, then freeze it, will it weigh the same?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Temperature | Pressure | Solid Amount | Liquid Amount | Gas Amount

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Heating from Frozen

Temperature starts at 0%, increase gradually

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Boiling Point

Temperature set to 90%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Pressure Squeeze

Pressure increased to 90%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Melting Point Threshold — The specific temperature where solid becomes liquid

Boiling Point Threshold — The specific temperature where liquid becomes gas

Particle Speed — How fast molecules move, which increases with temperature

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

When ice melts and water boils, does any matter disappear? Use evidence from your simulation to support your answers.

1. **Why does conservation of matter matter for science?**

2. **If you could see individual water molecules, what would you see happening during melting?**

3. **Why does high pressure turn gas back into liquid?**

4. **How is this model similar to what happens in the real water cycle?**

5. **What evidence would convince a skeptic that matter is not destroyed during boiling?**

8 STEM Challenge: Design a Phase Change Challenge

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YOUR ENGINEERING MISSION

Design an experiment that proves matter is conserved during a phase change.

THE SCENARIO

A younger student says 'water disappears when it boils.' Your team needs to design an experiment that PROVES the water didn't disappear — it just changed form.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU NEED FOR YOUR EXPERIMENT?
2. HOW WILL YOU MEASURE MASS BEFORE AND AFTER THE PHASE CHANGE?
3. HOW WILL YOU CAPTURE THE GAS TO PROVE IT IS STILL THE SAME SUBSTANCE?
4. WHAT DATA WILL YOU COLLECT TO PROVE CONSERVATION OF MATTER?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Materials Scientists study how substances behave under different temperatures and pressures. They design new materials for everything from phone screens to spacecraft heat shields. They earn $80,000-$130,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the phase change model, what happens when Temperature increases from 0% to 100%?

- **A.** Solid Amount increases because heat makes molecules pack tighter
- **B.** Solid Amount decreases, then Liquid Amount increases, then Gas Amount increases as matter changes state
- **C.** All three amounts increase because heat creates new matter
- **D.** Nothing changes because temperature does not affect the state of matter

2 What is the most important pattern the model reveals about total mass during phase changes?

- **A.** Total mass decreases because gas weighs less than liquid
- **B.** Total mass increases because heat adds weight to molecules
- **C.** Solid Amount + Liquid Amount + Gas Amount always equals the same total, proving matter is conserved
- **D.** Total mass changes unpredictably depending on pressure

3 What type of relationship exists between Temperature and Gas Amount?

- **A.** Negative: higher temperature reduces gas
- **B.** Positive: higher temperature increases gas as liquid evaporates into vapor
- **C.** No relationship: temperature only affects solids
- **D.** Negative: heat pushes gas molecules back into liquid

4 You weigh a sealed container of water, then boil it so all the water becomes steam. How does the weight compare?

- **A.** It weighs less because steam is lighter than water
- **B.** It weighs more because heat added energy and weight
- **C.** It weighs exactly the same because the same matter is inside, just in gas form
- **D.** It weighs zero because gas has no mass

5 In the model, what happens when Pressure is set to 90%?

- **A.** Gas Amount increases because pressure pushes molecules apart
- **B.** Gas Amount decreases because high pressure compresses gas molecules back toward liquid form
- **C.** Solid Amount decreases because pressure melts ice
- **D.** Nothing happens because pressure does not affect states of matter